

Summary

Software engineer and **Computer Science graduate** with hands-on experience building **production backend and full-stack systems** across product development, security, databases, testing, and cloud infrastructure. Experienced in taking features and production issues from **investigation through implementation, testing, deployment, and maintenance**, with a focus on **reliability and maintainability**. Strong in **Python, TypeScript, Next.js, Java, PostgreSQL, Supabase, Docker**, and cloud platforms.

Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE) Sept 2022 – July 2026
Vellore Institute of Technology, Chennai CGPA: 7.42

Skills

- Programming Languages:** Python, C++, Java, SQL, JavaScript, Bash.
- Tools & Frameworks:** Git, React, Next.js, Node.js, Express.js, RESTful APIs, FastAPI, Linux.
- Databases & Cloud:** MongoDB, PostgreSQL, Supabase, Amazon Web Services (AWS), Google Cloud (GCP), Docker.
- Core Concepts:** Data Structures & Algorithms, DBMS, CI/CD.

Experience

Hyr Mar 2026 – Present

Product Developer

- Resolved 100+ self-identified defects and 20+ client-reported issues** across Hyr’s recruiter platform, improving reliability across candidate management, interview scheduling, authentication, permissions, and result handling.
- Increased automated test coverage from **~1% to 95% across four services and 80% in another** by expanding **Vitest, Playwright, JUnit, and pytest** coverage for authentication, permissions, candidate operations, recording, and persistence workflows.
- Reduced interview recording failures and corrupted recordings from **effectively 100% to near-zero** in identified failure scenarios by rebuilding the **FFmpeg pipeline** with dual recorders, two-minute chunks, browser-side buffering, and verified uploads during live sessions.
- Hardened **tenant data isolation across five services** by implementing **Supabase row-level security, JWT/JWKS verification, and role-based access controls**, resolving authentication and authorization regressions through coordinated frontend, backend, and database changes.
- Improved production reliability and diagnosis across five services by implementing **Grafana and Sentry monitoring**, structured logging, metrics, distributed tracing, uptime checks, and Telegram alerts; also built **Hyr Live’s Chrome extension**, integrated **Hyr Agent APIs**, and independently developed **hyr.works** in Next.js after evaluating **16+ design prototypes**.

Apoliums Infotech India Pvt. Ltd. Dec 2025 – Jan 2026

Backend Engineering Intern

- Migrated legacy Node.js services to Golang (Gin) and optimized MySQL schemas, reducing latency and improving concurrent request handling.
- Designed and deployed RESTful backend services on GCP using Compute Engine and Cloud SQL, improving scalability and stability.

Research Intern May 2025 – July 2025

Centre for e-Automation Technologies (CeAT), VIT Chennai

- Developed a **CloudSim Plus** framework to evaluate **Hippopotamus Optimization (HO)** algorithm for VM placement in data centers.
- Enhanced simulation scalability by modularizing components, enabling reproducible testing across multiple datacenter configurations —**GitHub:** github.com/puneet-chandna/cloudsim-ho-research

Full Stack Developer Intern Dec 2024 – Feb 2025

Daira Edtech Pvt Limited

- Architected the Activity Management System** using Node.js and MongoDB, engineering RBAC with secure **JWT** flows to facilitate project lifecycles for faculty and external reviewers.
- Enhanced system reliability** and Student Progress tracking by refactoring legacy logic and implementing rigorous server-side testing, significantly reducing API errors and downtime.

Projects

Real-Time Dealer Gamma Exposure (GEX) Analysis for ODTE SPX Options Dec 2025 – Apr 2026

- Built a FastAPI engine for same-day expiry (ODTE) SPX options, processing 1,000+ contracts in under 1 ms using vectorized Black–Scholes–Merton Greeks, expiry normalization, and open-interest/implied-volatility filters.
- Built a Next.js exposure dashboard with cached REST APIs, 5-second WebSocket updates, and PostgreSQL storage; achieved sub-200 ms API latency and 99.5% WebSocket uptime.
- Used Welch’s t-test to identify higher realized volatility when net GEX fell below negative USD 1 billion.

Technologies: Python 3.13, FastAPI, NumPy, SciPy, Pydantic, PostgreSQL, Next.js, TypeScript, Zustand, WebSockets, Recharts

Github: github.com/puneet-chandna/ODTE-dealer-gamma

Key-Adaptive Parallel AES Encryption Engine Jan 2025 – Apr 2025

- Reduced encryption time by 51% on large payloads by engineering parallel AES-128 encryption using Python’s **ProcessPoolExecutor**.
- Increased throughput by 21.9% (362 KB/s vs. 297 KB/s) through a multi-core CTR-mode pipeline optimized across CPU cores.
- Designed SHA-512-seeded dynamic S-box generation with 49.95% avalanche effect, preserving AES-grade diffusion properties.

Technologies: Python, Multiprocessing, SHA-512, PBKDF2-HMAC-SHA256, Matplotlib, NumPy

Certifications

Professional Cloud Architect

Google Cloud | Credly Profile: credly.com/users/puneet-chandna.ea5ad5b2

Issued: Feb 2026

Positions of Responsibility

Outreach Lead, DAO Community:

- Grew community membership by 50% in 6 months through structured outreach, workshops, and collaborations with Web3 companies including Polygon and Huddle01.
- Secured partnerships for 15+ national hackathons and TOKEN2049 Singapore while leading a 35-member team to organize 5 large-scale hackathons, including Defy24, generating ₹5.5L+ in sponsorships and ₹3.5L+ in profit.

Blockchain Lead, Hackclub:

- Designed blockchain workshops, structured learning roadmaps, and mentorship programs to accelerate members’ Web3 skills.